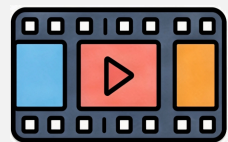
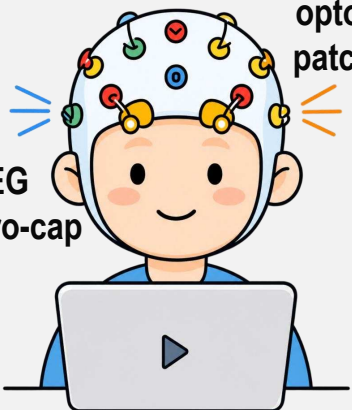


A

LEFT: INPUT SCENE



Familiar Video

fNIRS
optode
patches

New viewer

video time t

B

MIDDLE: TWO PARALLEL BRANCHES

Video-Time Prior (Low-Cost, Label-Derived)



Training participants

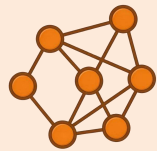
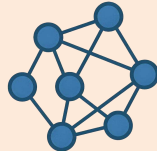
Fold-wise
median over
participantsRadius-3
within-video
smoothingVideo-Time
Prior
 $\bar{p}_{v,t}$

Held-out participant labels excluded

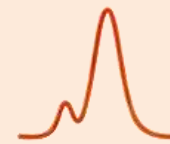
EEG-fNIRS Branch (Participant-Specific)



EEG (64ch)

fNIRS
(51ch $\times$ 6)1-sec
features
($t-1, t, t+1$)Graph Encoder
(EEG)Graph
Encoder
(α NRS)Cross-modal
Attention

Regressor

Physiological
Prediction
 $m_{s,v,t}$

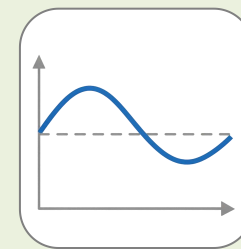
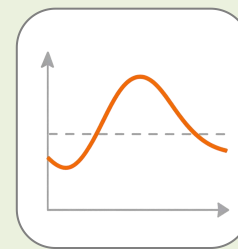
C

RIGHT: FUSION AND OUTPUT

Prior-Dominating
Fixed Fusion

$$\hat{y} = w \odot \bar{p} + (1 - w) \odot m$$

$$w = [0.99, 0.92]$$

Predicted affect
trajectory $\hat{y}_{s,v,t}$ Valence
over timeArousal
over time